

The Death of Evolution: A Documentary

Nicolas Jao

Dr. Elias Mateo Cruz, Chief Genetic Engineer, Philippine Genome Institute:

[The camera pans to a beautiful scenic array of all life on Earth, showcasing species from the Luzon rainforests, the Pacific Ocean, and the Atacama Desert.]

From the smallest insect to the largest whale, when you look at any living creature on Earth, you are witnessing the most recent product of billions of years of evolution.

For example, let's look at the complete lineage of the modern dog. When you look at a dog, you are really seeing the final culmination of one hundred billion generations of genetic relay over geologic eons, from the first multicellular eukaryotes to the early vertebrate fish, to the first land-based mammals, to the first carnivoraforms, to grey wolves, and finally to the domestic dog. If you plot the lineages of every species on Earth, it looks like one giant tree, and this is why we call it the "Tree of Life."

But what are the roots of this tree? Scientists have studied the genomes of thousands of species on Earth. They found similar sequences of DNA in every species they studied, suggesting that we all descend from one ancient species since the dawn of time. We refer to this species as LUCA, or the Last Universal Common Ancestor. The story of all life can be

explained by the rapid proliferation of LUCA into the splendid collection of species that has adapted to every niche corner of our world. We don't know what species LUCA was, if it was a simple or complex organism, or what kind of environment it lived in, but we know it lived at least 4.5 billion years ago. This was when Earth was still a hellish volcanic landscape, a planet still learning to walk and say its first words.

We are all genetically linked, and all life is no more than one big family. Families don't choose what their kids will look like, how smart they will be, or how healthy they will be. Two years ago, I advanced the field of biology to change that. I perfected CRISPR gene editing in humans.

CRISPR-Cas9 is the most revolutionary gene editing technology invented in the field of biology since its birth 300 years ago. It allows us to cut and paste any gene we want into the DNA sequence of any living organism, so we can choose what traits their next generations will have.

Today, I announce my gift to the world in the form of my firstborn son, the first designer baby. The advent of CRISPR proves to us that the trajectory of life can be precisely controlled by humans. For the first time in history, we don't have to be created as God intended. We can play God ourselves.

[A vibrant montage of species is shown again as music starts playing. There are timelapse animations of cells splitting, species evolving, more biomes such as rainforests, oceans, deserts, and finally, ancient apes in the savannah, which eventually leads to humans. Then it zooms out until the full spinning Earth is seen. The music reaches its climax. Then it stops, and the screen turns black.]

Malaya Science Films & Maple Insight Media

in association with

The Government of the Philippines

presents

THE DEATH OF EVOLUTION

*Warning from the International Documentary Association (IDA):
This film has been banned in 43 countries and is under ethical review by
world governments, including the US and Canadian governments.*

Elias Cruz:

[The camera shows Dr. Cruz's lab.]

Every day, as we walk about in our daily lives, an ancient and global war occurs all around us, a war that began when life first existed. Trillions of battles are waged every second on our skin, in our schools and offices, and in every corner of every environment on Earth. It is the war between bacteria and viruses.

Because these ancient creatures have been battling for billions of years, they have developed sophisticated weapons for their war with the aid of natural selection. One of these weapons is CRISPR-Cas.

Viruses are non-living entities that hijack a bacterial cell's machinery to copy more of themselves. They are very sneaky. At the molecular level, they're essentially just sequences of DNA. This allows them to blend in with other DNA. In other words, in the first days of the war, bacterial cells couldn't tell friend from foe.

Then they evolved. Hundreds of millions of years ago, when viruses were infecting bacteria, bacterial immune systems evolved to snip fragments of viral DNA from their dead enemies after surviving each battle. These fragments acted as "mugshots" to recognize their enemies if they ever fought them again. The bacteria stored them in a special region in their own genome. This special region, or list of mugshots, is called "clustered regularly interspaced short palindromic repeats," or a CRISPR array.

The tide of war was turning in favour of bacteria. Now they had a superweapon to expose the disguises of their enemies. They did this by transcribing viral DNA from their CRISPR array into RNA molecules, which were then attached to Cas9 proteins, their soldiers. These RNA molecules acted as guides for the soldiers. The soldiers could now do a border check on the walls of bacterial cells, and if the RNA molecule was a perfect match with the invader's DNA, Cas9 would cut it—because it meant the virus was back.

The biggest breakthrough was when we discovered we could program this system ourselves.

It was simple. All we had to do was make our own RNA guide, give it to Cas9 along with a "package," the gene we wanted it to replace, and it would go find the gene in the DNA sequence, cut it, and paste the gene we wanted instead, all by itself. It's so precise, it's like editing text in a document. We had stolen bacteria's superweapon to make our own.

What do I want to achieve with this superweapon?

Freedom from the Tree of Life.

[Footage of Olympian athletes racing each other, their sweat glistening as their winding muscles allow them to glide over the track. It is followed by footage of basketball players, professors teaching lectures on chalkboards, and fashion models on catwalks.]

Every person, no matter how much they end up achieving in life, is ultimately limited by their genes. According to general scientific consensus, it is true that among humans, there are no superior genes and we are all genetically similar. However, the slight variation that exists is still enough to ensure some people will never become professional athletes, solve the world's toughest math problems, or become a superstar based on the features of their face. But aside from those arguably superficial things, I aim to focus on something deeper. Although all of us are 99.9% genetically alike, this still does not solve one of the greatest issues of our globalized world: discrimination.

Discrimination is a silent cancer on society. It is based on phenotypes—how one person looks, regardless of their overall genetic makeup. This means the problem is powered by just a handful of genes—the ones that create what you look like. In some stages, it takes the form of blatant aggression shown at you—this is no problem, because tumours that reveal themselves loudly are easily spotted and taken out. But more often than not it occurs in its quietest stages, in the tiniest influences in the subconsciouses of the people you interact with. Here, it is completely invisible—and therefore at its deadliest.

Whether we like it or not, the appearance of a person affects the way we treat them. But almost conversely, the appearance of ourselves affects the possibilities we allow our identities to reach as well. Why are some ethnicities more likely to commit crimes, more likely to earn higher degrees, or more likely to win matches on dating apps? I don't believe social change will ever fix the tribalism inherent in human nature because the problem is not a social one. It's a scientific one.

For millennia, tribalism in humans evolved because small, cohesive groups offered survival and reproductive advantages, such as protection, resource sharing, and cooperative care. You can't control it—it's ingrained in our genes to form groups, to identify similarities with some over others, and to form group loyalties to survive. How can we socially solve something in our genes? A scientific problem requires a scientific solution. With CRISPR, we can finally do something about it.

My goal is to provide people of all backgrounds a way to let their children escape the curse of discrimination. I want to charge my CRISPR procedure—which I call *Genaris*—to people by class: free for parents in systemic poverty, and a high price for parents in the upper classes, with the highest prices for the richest. In this way, the rich will fund the poor. All customers will be able to genetically edit their babies and give them whatever features they want, ensuring their children can have an advantage in their lives.

Regardless of your personal take on race, it *will* affect you. Even though I'm back in the Philippines, I've lived in America since I was eleven years old. Every immigrant who comes to America knows something about being an outsider trying hard to become American. For some children, they also know something about being the third or fourth generation of their family to have been born here, yet still have their "Americanness" questioned. But if you are part of the fair-skinned majority, you will never be questioned. My goal is simply to eliminate the long-standing inequality between darker-skinned people and lighter-skinned people. Only then will we be one step closer to a perfect society.

Because I am human, I myself am under the tribalist tendencies I criticize. Therefore, in my heart, I must start with my own people. For far too long, the Filipino people have suffered from a brainwashing called "colonial mentality." We prefer foreign brands to our own. We prefer paler skin over darker skin. We believe one of the best things we can do for our lives is to get out of our country, rather than help it from within.

Filipinos love escaping. With Genaris, we can escape our skin as well. Now we can truly be free.

Narration:

[Stock footage showing the construction of a twenty-meter-tall bronze statue of a Filipino warrior-king, perched on a white pedestal and surrounded by a landscaped garden. The statue is depicted as a muscular figure wearing a native loincloth called a bahag, wielding a traditional curved sword called a kampilan in one hand and a wooden shield called a kalasag in the other. His towering presence generates an aura of respect and readiness for battle.]

We are on Mactan Island, a small island near Cebu in the Visayas. It has been said that a great battle occurred here, symbolizing the first blood spilled between indigenous Filipinos and Europeans. On April 27, 1521, Ferdinand Magellan and his Spanish troops, allied with Rajah Humabon, a Cebu chieftain, invaded the island to settle a dispute with a local ruler.

The name of this local ruler means nothing to most people in the West. In fact, they might know it better as the Filipino name for a fish, called a "grouper" in English. But to Filipinos, this name is a national hero and has been taught in schools as early as Grade 5. His name was Lapu-Lapu.

[The screen shows a Baroque-style animation of various historical events. The Spanish King's court, the issuing of decrees, the dispatching of explorers, and the loading of ships with cargo. It cuts to ships sailing, as if flat objects in a painting in motion, with swirling blue tides bustling up and down.]

For weeks since his arrival to the archipelago, Magellan's ships had been anchoring at island after island, forming alliances with local kingdoms called *barangays*, and baptizing their leaders into Christianity. Most of them, including Humabon—the first chieftain Magellan met—wanted political power and resources from the foreigners, and they submitted to the Spanish Empire immediately. All around the islands, barangay after barangay joined the empire, following the decree set by Magellan and Humabon. In the midst of subjugation shone a ray of hope: Lapu-Lapu and his tribe were the only ones who resisted.

Enraged, Magellan ordered his fleet to conquer them. Due to low tide and coral reefs, they had to abandon their ships far offshore and wade through chest-deep waters toward the beach. Helmets and armour weighed them down and made them vulnerable as they carried their firearms. Little did they know, in the dense jungles behind the beach, a fierce army of 1500 warriors was ready and well-positioned on vantage points, ready to pounce on them.

The battle was vicious. Lapu-Lapu ordered attacks using poisoned arrows, fire-hardened bamboo spears, and blades called *kampilans*. They focused on hitting the Spaniards' unarmoured legs, using hit-and-run tactics and flanking movements.

Magellan ordered the burning of nearby houses to provoke open combat. But rather than intimidate, it only enraged the defenders and aided their flanking advances. Spanish forces began to falter.

The decisive moment came when Magellan led a covering maneuver to regroup his men. Lapu-Lapu's warriors seized the chance, and in close combat, Magellan was struck by poisoned arrows and spears and was killed in shallow water. With their leader down, the Spanish troops collapsed and retreated to their boats under heavy pressure.

This is the earliest recorded event of a native Filipino tribe successfully defeating a colonial European power.

To many Filipinos, this is the true start of their country. Its impact on history was undeniable. It gave peace to the islands by delaying the specter of colonization by at least a few decades. It was a full 44 years before the first Spanish settlement on the islands was built, and another 333 years before the Spanish were finally ousted by the Philippine Revolution, creating the modern state we know today.

Magellan may be known for leading the first expedition in history to circumnavigate the globe, but he did not personally complete the journey. The only reason historians know of this battle today is from the accounts of the Spanish survivors. Today, it is mostly Filipinos who know this story.

But ever since the arrival of Genaris, Mara Tan was determined to make the whole world know.

[The camera shows old footage of the completed statue with a woman in front of it. She reverently looks up at it, her hands crossed behind her back. No one in the crowd of journalists and reporters behind her knows what she's thinking.]

Dr. Mara Tan, a PhD in computer science who specializes in AI video generation, has made it her life's mission to take down Elias Cruz's CRISPR movement. As a native Filipino who studied in Canada, she claims this is a mission to restore the dignity of the Filipino people.

The Philippine government has made its official stance clear to the world that it condemns Elias Cruz, but there is evidence of an internal debate about the technology, splitting it into two sides. Efforts to ban the technology within the country have been ineffective so far, and rumours have spread that Elias Cruz has paid corrupt government officials to allow him to import the necessary lab equipment from the United States. A booming black market has organized in the shadows, providing Elias Cruz's operations sanctuary from law enforcement. As a result, Mara Tan has refused the aid of the Philippine government and has chosen to act privately. Most of her funding comes from charity from supporters around the world.

Her goal is to use AI, a cheap way to generate hyperrealistic videos, to spread awareness of the Philippines' colonial history to the world. Her AI videos have gone viral on the internet for their brutal portrayals of colonizer atrocities on the Filipinos, as well as their hopeful and optimistic representations of traditional Filipino culture. Many of her critics allege she is involved in a nationalistic propaganda stunt, fuelled by an unjustified hereditary sentiment to sway public opinion against Elias Cruz's revolutionary breakthrough.

On this hot summer day, Mara Tan faces a crowd of journalists as she explains the story of how visiting this statue of Lapu-Lapu inspired her as a kid. She holds a speech at a podium, taking us on a narrative journey of her life. She starts with her interest in her country's history when she was young, to her interest in AI for a career as she grew older, and finally to the monumental day she decided to join the war against "colonial mentality," like many other Filipino heroes before her.

Much of the world is ready to move on from past atrocities, framing colonialism as just another part of our unchangeable history, believing that the only solution is to forget and keep going forward. But not Mara Tan. She believes ignoring the horrors of the past is a great injustice to those who suffered, that all contemporary issues could be explained by history, and that it is a moral duty to remember victims of the past, no matter how far we move forward. With many people around the world signing up for Elias Cruz's CRISPR treatment, dreaming of paler skin for their children, she believes people today have forgotten all of this.

[Mara Tan leans into the mic on her podium and says the last words of her speech. Cameras snap with white flashes as she speaks.]

“Dr. Cruz and I believe one thing in common: The genes of the past carry on in the genes today. These people contained the original genes that we today would call “Filipino.” The modern Filipino people are the direct descendants of these indigenous tribal kingdoms. In a way, we share a connection with these people, and no matter what we do or say, no matter how much we believe we can escape it, we carry their legacy in every moment we’re alive.

“Our ancestors died fighting colonization because they loved our people and had hope for our people, and I hope you can feel the love of a million ancestors whenever you start feeling ashamed of your Filipino identity.”

Dr. Rafael Santos, William Henry Scott Chair in World and Philippine History, University of the Philippines Diliman:

José Rizal, the greatest national hero of the Philippines, wrote a book that kickstarted the Philippine Revolution against the Spanish Empire. He called it *Noli Me Tangere*, Latin for “Touch me not,” because at the time, the Philippines was suffering from a social cancer from colonialism so severe, it could not be touched or addressed without causing further pain and harm. Over a hundred years later, into the new millennium, after the Philippines settled into being its own independent republic, Filipino psychologists have studied the Filipino population and have reached one definitive conclusion: That social cancer still exists today.

When I was in London for an international history conference to present my research on Southeast Asian colonial trade networks, I took the London subway to get to work. One day, I overheard a conversation on the platform between two young Filipino friends. The girl, who had a precolonial Filipino name, was expressing to the boy, who had a Spanish name, how much she hated her name and wished she had a Spanish name too. Her words struck me with a deep, unexpected sadness.

Look no further than the Philippines itself. As a professor who teaches the youth, I witness firsthand the reality of our national identity crisis. There are self-hating Filipinos who hate their roots due to being exposed to dominating foreign cultures in media, or by the pervasive corruption they see in all aspects of our society, convincing themselves that it is the only thing we are destined for. Some totally abandon their Asian identity entirely and embrace being “whitewashed,” or even strive for it.

The term experts have given it today is “colonial mentality.” It’s an ethnic and cultural inferiority complex, often systemically instilled in our youth. It’s seeing a Facebook post about a mother with a *hafu* baby, overjoyed about how the baby doesn’t look very Filipino. It’s using papaya soap to look “civilized and beautiful.” It’s me speaking exclusively in English for this interview, without a single Tagalog word, so it can reach the international audience more effectively.

While it is saddening, I understand the inclinations behind it. Eventually, we may reach a point where we sacrifice our weakening culture for progress. Then, just endure the identity crisis until we hit the “greener” pasture. Many Filipino kids, most especially the diaspora, are in danger of abandoning their roots. It only gets worse when you deny yourself a sense of origin and place in this world. Destroying the origin for a new one might be the only way out, and that’s what Elias Cruz sees.

Vance Hao Tan, husband of Mara Tan:

[Vance Tan looks to be in his early fifties. He looks gaunt and tired, but in his eyes remains a hopeful light reminiscent of the kindness and hospitality Filipinos are known for. He cracks a lame joke, and the crew laughs.]

Although I consider myself culturally Filipino, also fully fluent in Tagalog, my background is Chinese. I used to call my grandmother *a-má* (阿媽), and my grandfather *kong-kong* (俺公), both on the paternal side. My mother had a little Chinese blood but was mostly Filipino, so I called my grandparents on her side the traditional Tagalog words for grandfather and grandmother, *lolo* and *lola*.

I was very close to my *a-má* and *kong-kong*. I have memories playing around the spiral staircase of their apartment in Binondo with my brothers, or playing with these huge blue jugs of water for the water dispenser. The water in the Philippines is not very clean, so we always have to drink from these plastic jugs. I also remember studying the lines of ants that formed trails in our kitchen, or the little dark brown geckos that scaled the walls everywhere. They were about the size of my thumb.

Under the sweet, hazy guise of nostalgia, this was the apartment that I grew up in, and I always look back on it very fondly. I knew it as my grandparents’ home, but since my family lived there, it was also mine. It was a place that I was always welcome, a place where lots of relatives gathered and played mahjong—fat uncles and gossiping aunts, more cousins than I can count, and mountains of *lechon* and *pancit*.

My grandparents died before I got to high school, my *kong-kong* first, and then my *a-má*. After that, we moved to Greenhills. On her hospital bed, my *a-má* said, “I want to see you get married.”

“What kind of girl should I marry?” I asked her.

“Marry someone who always feels like home.”

For years after their deaths, I felt I was losing my connection to this feeling of “home.” My *a-má* made me take Mandarin lessons since kindergarten, but my mom stopped them because she didn’t think they would be that useful—especially since Chinese Filipinos speak Hokkien,

not Mandarin. My brothers grew up and didn't want to play as often, and then they studied abroad. Even the family gatherings that peppered all stages of my childhood gradually shifted from constant to occasional.

Mara brought that feeling of "home" back into my life. That's why I fell in love with her. She was very close to my family and helped us a lot when we were dating. She helped my father down the stairs and spent time in the kitchen with my mother, learning her recipes for black chicken soup and *lumpiang sariwa*. I was shocked, but also touched, by her tears when my mother died. She didn't know much about Chinese Filipino traditions, but she tried her best to learn: the correct number of joss sticks, the amount of money to put in each red envelope, and how to fold the money rolls that we'd send with them in the afterlife.

Every detail mattered to her so much that it made me laugh. She'd make a fuss about doing something wrong, and when I'd ask her what it was, it would be something minuscule that I had never even thought about. I'd see her putting more joss sticks in the pot because she was worried that too few might offend my dead grandparents. I'd walk past the kitchen and watch her measure the length of the red envelope just so the flap wouldn't wrinkle. All the while muttering in a way that made me giggle, because she looked both serious and nervous, like she was bracing for one tiny mistake.

We were caught out in the rain once, when there was a storm. We held onto our umbrellas tightly but still got soaked. The kids on the street continued to play basketball, even as the street got flooded, and when their ball bounced near us, we'd get splashed with murky brown water. We'd laugh. We didn't try to run, didn't even try to find shelter. We wanted to experience the rain along with those kids.

"Why are you so scared of getting my family's traditions wrong?" I teased.

She thought about it.

"We think the living are all who matter on this Earth, but we would be nothing without the people that came before us," she said. "I get so sad when people fail to preserve the fire of their families just because they don't know how, and are left only with ashes. What they end up with is a sense of a loss of home, as if they've forever been disconnected from a part of who they are. Traditions are how we keep the votes of the dead alive in this democracy of human existence. So for yours,"—she smiled and jabbed her index finger at my chest—"I want to make sure I get them right."

Mara made it her mission to take in the stories of strangers and bring them to life, and this was why.

Maria Galang, Filipino elementary school teacher:

This Mara Tan is crazy, *sira sa ulo*. *Hindi niya alam na* Dr. Cruz is also helping dumb people, ugly people, and diseased people. It's not just about race; Mara Tan is just making it all about race and colonialism. Skin-lightening is just one of the things Dr. Cruz offers, *pero* he's also offering so much more. *Sayang*, if we hold ourselves back from all the things we can do with this technology just because of a bunch of snowflakes and their queen.

Antonio Dela Peña, Filipino Canadian immigrant:

The Western colonial powers civilized us and advanced our societies. Before, we were a Stone Age society living off hunting and gathering, not out of a love for our subsistence culture, but because we had no choice. This romantic image that Dr. Tan has about finding our precolonial culture valuable is nothing but an illusion for fools. Arguably, she's right that Dr. Elias's motivation to use CRISPR to give Filipinos lighter skin is more social than scientific, but when you consider the real social effects of looks in society, the decision does become practical. If better looks improve the quality of our lives, then so be it. At the end of the day, we're all just trying to get by.

Vance Tan:

Mara spent a long time thinking of what to name her AI model. We went through a list of names on a sheet of paper, but each one failed to leave her fully satisfied.

"Your AI can do a lot of things," I said. "But what is its *purpose*? If you took away everything about it, what would be the last thing you'd have to take before it stopped being itself?"

In that moment, I saw a fire ignite in her eyes. There has always been one thing I have known about my wife. Growing up, Mara felt that the history textbooks of our nation were frustratingly inadequate. She believed in a tragic lack of media representation of the Filipino history, particularly on the world stage. For far too long, we've understood the story of our people primarily under a Western perspective first and how we relate to foreigners, but never focused on just ourselves. Dozens of generations have lived and died on this land with nothing more than a footnote in an American history textbook to show for it.

"*Gunita*," she said. "I want the world to remember."

We ended up calling the model GunitAI. It would be a revolutionary type of model, the kind that would change the world. It would specialize in transforming language prompts into realistic videos, turning stories of the imagination into a form people can absorb and digest in

the blink of an eye. All she would have to do was type out a story using words, send it to GunitAI, and we'd see the final result.

This allowed her to mass-produce them, and she aimed to use it to generate horrifyingly accurate videos displaying the violence, rape, cultural erasure, and other atrocities committed by the Spanish during their colonial rule over the Philippines, followed by the Americans in the years after. She wanted to show the world the real stories, both written and oral, of families who remembered them from their ancestors.

With remarkable speed, our critics came at us.

Why are you stirring up things from the past and ruining our relationship with the West? the Filipino ones said. *Many of your videos criticize the things America has done. But we need America to protect our waters against China, and America has been so good for our economy.*

There is no conceivable circumstance in which your videos deserve an actionable response, the American and Spanish governments said. *They were made using an AI and cannot be legally used in any court or applied as evidence. They're fiction.*

Day after day, people berated her work, but also every detail about her life. They criticized her decision to study abroad in a Western country instead of at home, which didn't sit right with some Filipinos. They said she had a colonial mentality as well because she wouldn't marry a "real" native Filipino, but rather one with Chinese ancestry instead. She was staunchly anticolonial, but they used that as an excuse to call her fiercely nationalistic. She was digging up crimes of the past, so they blamed her lack of credibility to defend their status quo. She was tired of other people making the rules, so she started breaking them herself.

Dr. Aleksander Kowalczyk, Professor of Computer Science, David R. Cheriton School at the University of Waterloo:

The first attempts at artificial intelligence were modelled after the human brain. It was fitting that we gave respect to it. After all, it was the source of intelligence itself—it seemed obvious that mimicking its structure would create what we wanted. First, we looked at neurons firing electrical signals at synapses in our brains. Then we thought digitally simulating this could create non-sentient structures that gave us correct answers by learning from feedback. We called them neural networks.

Today, neural networks are nowhere near as sophisticated as the brain. But they still get the job done pretending to be.

This is an oversimplification, but the truth is, even computer scientists do not fully understand all the internal mechanisms by which neural networks arrive at their outputs. It's a black box—you feed it something, it spouts something out. The math behind them is well-understood—the forward pass, the backpropagation, and the optimization algorithms. But

what remains mysterious are things such as why specific configurations of weights lead to surprisingly effective solutions, how networks find generalizable patterns rather than memorizing data, or which internal representations correspond to meaningful concepts, like “cat-ness” or “justice.” We built the engine, understand how to tune it, and can drive the car—but we don’t fully understand how it chooses the best route through the fog.

To think that because we don’t fully know how they arrive at their answer, they could produce anything, is scary. Their creations have unlimited potential, whether humans are involved or not.

Mary Winters:

I was one of the first to sign up when Mara Tan’s AI video service was announced to the world. Even so, the waitlist was long, especially since she prioritized the stories of Filipinos first before Filipino Americans. I wouldn’t get my appointment with her until a month later.

Most people today know Mara Tan’s “positive” videos as well as her “negative” ones. But they don’t remember what it was like before the positive ones. When the first videos came out, they were *horrifying*. We saw detailed videos of Spanish friars raping Filipino children, with the sound of their screams so realistic, it felt as if they were crying out for help through the screen. We saw American troops bombard groups of Filipino soldiers with artillery, and then their disconnected legs, arms, and heads would go scattering through the air, and the American soldiers would laugh or make bets on who could score more at once. It seemed like Mara Tan’s goal was to remind us that colonization was not just conquest, but cruelty etched into generations. The world wasn’t ready to watch any of it, and sometimes the videos would be banned or flagged for graphic content. But for those who watched them, they were enraged, and they were ready to give her lots of support.

I wish I could tell everyone that it doesn’t have to be this way. It was only when I got my turn to produce a video of my great-grandfather that the other side of her videos’ potential was unlocked. I am the one responsible for Mara Tan’s “hopeful” videos.

I live in Santa Cruz County, roughly 75 miles south of San Francisco. Last week was my 46th birthday. When people take a look at me today, they would never guess that I have Filipino roots. That ancestry is less and less visible with each generation in my family, and I fear that if I don’t speak up about the stories of my great-grandfather and this area’s first Filipino immigrants, their memory could be lost forever.

My great-grandfather was born on April 17, 1897. This was a time after the revolution against the Spanish, when the Philippines falsely believed the Americans were its allies. As a result, my great-grandfather was part of the *Manong Generation*, the first major wave of Filipino immigrants into the United States.

The Manong Generation were mostly young, single men who left the Philippines looking for work. They mostly crossed the Pacific and settled here, in the Pajaro Valley of California. In 1916, my grandfather left on a boat at 19 years old, came here, and spent 11-hour days picking strawberries and hauling irrigation pipes to survive. Even though he and the other Filipinos he came with were paid less than both white and Mexican workers at the time, he still often found a way to send a portion of his wages back to his family across the Pacific. I will call him *Lolo sa tubod*, the Tagalog way to say great-grandfather.

As a little girl, I heard stories of him from my grandmother, his only daughter. She was the last living connection who knew him. I knew that at the time, the white Americans held a great prejudice against foreign labourers, especially Filipinos, who they believed were taking all their jobs for less pay. Since the US only allowed Filipino men to come for work, these Filipinos were unable to meet Filipino women and had no choice but to seek companionship with the local white women. The whites saw this as a threat to their society, and tensions began to rise.

My *lolo sa tubod* was one of those men who secretly married a white woman, which was illegal at the time. In 1918, my grandmother was born, and my *lolo sa tubod* did his best to raise her at a distance.

“I was a happy child, and I loved my father,” my grandmother said. “At the time, I was too young to know about the social ecosystem of my community. I was innocent against every nuance, every anti-Filipino sentiment of the Americans. I only remembered Sunday mornings picking strawberries with him and him teaching me Tagalog through *kundimans*—traditional Filipino folk songs. I didn’t fully understand that kids bullied me at school because I was half-Filipino, or why he and my mother were very careful not to be seen together in public.”

Then in 1930, tensions reached a climax when reports were spread of street carnivals with Filipino men walking with their white female companions, and taxi dance halls of Filipino men dancing with white women. The whites were outraged, and soon they started organizing mobs to attack Filipinos all over the county.

Riots started breaking out. In Watsonville, a few miles south of where I live, white men were ganging up on Filipinos, gathering clubs and weapons, dragging them out of their homes and beating them almost to death, and burning down any place associated with them. The Watsonville police did their best to guard as many Filipinos as they could in the City Council’s chamber. At the Pajaro River Bridge, Filipino men and white men were facing off to resolve the matter once and for all.

“Your great-grandfather died during those riots,” my grandmother said.

“How?” I asked.

“He was one of those men who fought on that bridge. He suffered a heavy beating, and he didn’t survive. The white men threw him over the bridge and he fell into the river. I was only 12 years old when he died.”

She wrote down everything she could remember about him. When my mother was born, she did her best to transfer the memories to her through stories. For years, she held onto the one black and white photo she had of him: him and his wife at a dance hall, all dressed up, staring at the camera with expressions both solemn and happy at the same time.

I asked Dr. Tan to make me a video of my *lolo sa tubod*’s death. I knew that’s what she wanted—stories of violence and hatred to inspire the world to action. I gave her the photo and we watched GunitAI generate it in real time, witnessing the mysterious inner workings of the neural network do its job to complete the memory.

The final output was finished. We saw a bloody fight, and then three men from the white mob pinned *Lolo sa tubod* against the railings of the bridge, preparing to throw him into the river. His eyes were pinched shut from beatings, he was missing teeth, and blood ran down his forehead. I winced.

“You can look away if you want to,” Dr. Tan told me.

I shook my head. “No, I want to watch.”

Then something unexpected happened. My *lolo sa tubod* started singing a proud tune, and Dr. Tan and I were stunned. His voice was raspy and weak, but the AI lowered the noise of the chaos around him so he could be heard.

He was singing the lyrics to the Philippine national anthem. At first, we thought something was wrong, but later we theorized perhaps GunitAI was thinking he would be patriotic in his last moments.

As I watched my *lolo sa tubod*’s last moments before death, I was overwhelmed with emotion. Through the screen, time slowed down, and his song pulled me in so deeply I couldn’t help but sing along:

*Lupa ng araw, ng luwalhati’t pagsinta,
buhay ay langit sa piling mo,
Aming ligaya, na pag may mang-aapi,
ang mamatay nang dahil sa iyo.*

*Beautiful land of love, o land of light,
In thine embrace ‘tis rapture to lie,
But it is glory ever, when thou art wronged,
For us, thy sons to suffer and die.*

I was beginning to learn that reality is not in what we see or hear, but in what we feel.

Loretta Lasangan, nurse:

Here in Manila, we've already started offering Dr. Elias's CRISPR treatment as "fair-skin bundles" to pregnant women, which includes other selectable genes they want for their child. We market it as "correcting genetic defects," and then our customers are secretly told they can make designer babies. Before we give the bundles, we do a background check on their class to ensure we charge them the correct amount. It's been very good business so far. It makes the clinic staff feel good about giving poorer people a chance to rise up in the social hierarchy, and at the same time, we are taking money from the ultra-rich. We've been in contact with other clinics in the country, and many of them are starting to do this too.

I became a nurse to help people and change their lives, but doing that is so hard with our corrupt government. For once, I feel like I'm actually making a difference.

Alfonso Baraquel, bank manager:

I spoke about it once at my mother's birthday party. We are very divided on this topic. Some of the *titas* gossip that they want it for their kids, but some of the *lolos* say God created us as intended, and tampering with that is a sin. The kids are mostly against it, as they had been educated about colonialism in school very strongly. My brother's eldest kids explain that there are huge resistance movements on their school campuses to reject designer babies, that they want to be free from colonial trauma, not revive it. Some of their cousins are quiet about it, and I suspect they support the idea more than they want to publicly admit.

My wife and I don't have children yet. So, inevitably, the choice will fall on us. At first, editing our children was a hard no for me. But then I started fearing what might happen to my children in the future if I didn't. My wife started supporting these thoughts.

If, all of a sudden, everyone was doing it, it was certain our children would be put at a disadvantage. They'd struggle to get dates, struggle to get scholarships, or struggle to even make friends. They might get bullied for having darker skin. Think of the escalating competition of schools in China: if one person in the class studies harder, everyone else has to match them to keep up. If someone else studies even harder, then everyone else has to match that person again. This is why all of them have hours of after-school classes and their workforce has the infamous 996. It goes up until everyone is exhausted, and no one gets the jobs anyway because there are diminishing returns.

The thing is, with CRISPR, you have to make your decision at your children's birth. It's not easy. You decide what the rest of their life will be like, and you only get once chance.

Erica Cuizon, college student:

I was part of my school's protest against Elias Cruz's CRISPR movement.

At first, we were all in on it. We hated the idea of editing genes because we were deeply scared of what it could mean for society. It could create another pandemic of genetic superiority—you know, the kind that was widespread in Nazi Germany, the kind that caused millions of deaths in the past. So we wanted an outright ban on anything to do with it.

Then the first pictures of grown up CRISPR babies came out. A couple of us gasped. I remember looking at one—he was smiling at the camera, and I said *Oh my God*—my heart completely stopped. Over the next few days, I started fantasizing about being with someone that beautiful.

The other girls made an effort to condemn the pictures, but we all saw the same thing. I know some of them felt what I did, too.

I felt extremely guilty about it. I thought about intentionally dating an ugly person, just for the sake of rebelling, and for a while I thought that was a good idea. Later, I realized that was stupid, because why would someone want a partner who was only with them out of pity? And why would someone choose an ugly partner just to signal their virtues? It was transactional, and I knew that wasn't the way.

Then I thought, didn't this transactionality apply to most relationships? We all go for someone we think is in our "league." If we all had the chance without feeling insecure or guilty about it, wouldn't we go for someone more attractive? At least Elias Cruz was levelling the playing field for everyone, allowing all of us, no matter what class or race, to be attractive.

I thought these ideas myself. But when other girls started reasoning similar things, and as they started exploring Elias Cruz's side, suddenly the thoughts made me sick. I started distancing myself from the whole protest because I didn't want to think about any of it anymore—even from my friends who were still purely against it.

"Where are you, Erica?" my friends asked. "We need your help. More people are signing up for CRISPR. Ignoring the issue doesn't help anyone."

It was a little selfish, but I just needed time to think. I haven't been to a protest on campus for a few months now. I don't know how well they're still doing.

Mary Winters:

The second time I had an appointment with Mara, I requested something new.

I asked her if she could make me a video of my *lolo sa tuhod* and my grandmother surviving and being happy.

"I've never done this before," she said.

I encouraged her to try. She spent a day training the AI on new data, and she promised she would email me the result. I received the email a few days later. When I got the email, I opened the video file and saw a version of reality that filled me with a great sorrow, but also an unspeakable happiness.

Instead of creating a video with remarkable realism, the AI rendered small moments that never happened but should have, moments from a parallel life.

In the video, I see *Lolo sa tubod*, whole and unbroken, walking hand in hand with his daughter—my grandmother—along the riverbank. His clothes are still the same work overalls, but they're clean, sun-faded from years of honest labour, not violence.

He's showing my grandmother how to pluck ripe strawberries without crushing them. Then the two of them are laughing as they ride home on a worn-out bicycle with one broken pedal. Then he's sitting on the porch of a small wooden house, humming a tune while his wife braids their daughter's hair beside him.

As night falls, the perspective shifts: I see the Pajaro River Bridge again, but it's rebuilt in concrete, with cars passing peacefully. Generations later, a woman—my mother—stands there, placing flowers in the water. The video quietly merges the timelines, and across the river, faint silhouettes of my *lolo sa tubod* and grandmother wave, smiling.

Text fades onto the screen, generated by the AI in stylized Tagalog script:

"Hindi lahat ng alaala ay dapat masaktan. May ilan na dapat gamutin."

("Not every memory must hurt. Some are meant to heal.")

My mother has had issues with her heart for years now. Two weeks ago, I visited her in her hospital bed to show her the video. She smiled so much, it made me realize I've never seen her that happy before.

She passed away last night. I'm happy that in her last moments, she was able to see an alternate history where her grandfather never died in the riots. We both know it's fake, made by an AI, but we didn't care. For a few minutes, my great-grandfather was alive.

AI videos could be used for harm, but they can also be used for a lot of good. Dr. Tan learned this, and that's when she was inspired to start using the AI videos as a tool for positivity, not just negativity. She started believing that a mix of both was the best path to getting more Filipinos on her side.

"I want to thank you," she told me. "I'm going to call for all Filipino historians to focus on digging up good stories from the past now."

Narration:

[A sequence of scenes shows Filipino consumers using various beauty products in shopping malls. There are cuts to famous Filipino actors and singers.]

The preference for lighter skin in the Philippines is deeply tied to colonial history, where lighter skin was associated with Spanish and American colonizers, wealth, and social status.

This has led to widespread colorism, where lighter skin is often seen as more desirable, influencing beauty standards, media representation, and even employment opportunities.

The Philippines is a large consumer of skin-whitening products in Asia, with a market projected to grow steadily through 2025 and beyond. Products range from creams, soaps, and lotions to oral supplements and injectables, sold both online and offline.

The market is driven by high demand, especially among younger consumers and urban populations, making it a lucrative sector for both local and international brands. Many Filipinos face social pressure to conform to lighter skin ideals, which are reinforced by advertising, celebrity culture from the West, and social media. This has led to widespread use of whitening products, sometimes without full awareness of unregulated potential health risks.

Rafael Santos:

While it is true that European colonization amplified the lighter skin bias, Mara Tan is making a mistake by believing this was the true origin of it. The lighter skin bias has existed in Filipino society long before the colonizers came.

The Visayans had a tradition called *binukot*, where young women, often daughters of rulers, were isolated from a young age from public view until marriage. Intended to preserve their beauty, the seclusion protected them from the sun, resulting in fair skin. It became a highly valued trait, which was expected to increase their worth in the eyes of potential suitors.

Charles Darwin noted that even in populations with little European contact, lighter skin was often admired. This suggests a cross-cultural convergence in beauty standards, possibly rooted in evolutionary psychology. Whatever the case, Mara Tan is wrong to assume the global lighter skin bias as a sentiment of colonial mentality, when it is much more complex than that. Of course, I don't expect someone with her character to understand this nuance—she seems to be fevered with a drive to take down those in power based on her personal agenda and how she was raised. She suffers from the inescapable effects of culture and propaganda, as all of us do.

Gia Reyes:

Maple Insight Media Reporter: On behalf of MIM, thank you for volunteering to speak up about Dr. Mara Rosario Tan. Can you start with how you knew her?

Gia: Yes, I can. I first met Mara at university. We were both writing our computer science doctorates at Waterloo, and we were practically inseparable at the time. Then we graduated, and ever since she left to go back home, we've drifted apart. But we still chatted and caught up once in a while. I don't know exactly what she was thinking as she poured her efforts into the campaign against Elias Cruz, but I can at least tell you what she was like at Waterloo.

From our time in university, Mara was already a very empathetic person. She was the type of person people were jealous of, the type they aspired to be. She excelled in all her classes, was extremely nice to everybody, and all the boys had a huge crush on her.

When we weren't studying, she focused all of her time on trying to make a difference. She set up a fundraising booth in the Student Life Centre for disaster relief when Mount Pinatubo erupted. She advocated for the rejection of US military bases in the Philippines when anti-colonial sentiment was at an all-time high. She constantly needed my help putting up posters to call attention to the Muslim separatist groups in Mindanao, who were fighting for the freedoms of the Moro people. That one was particularly controversial and got her in a lot of trouble, especially with our other Filipino friends.

As we went through the years, she gained more interest in Filipino history. She read more textbooks and consulted more professors. She'd complain about the colonial architecture of the San Agustin Church, or talk about the origins of menudo. I noticed many of our conversations turned into long discussions about Spanish and American colonialism. Much of it was new to me.

I was born in the Philippines, but I moved to Canada when I was seven. At the time, I couldn't understand what made Mara so interested in this topic. Sometimes I found her antics ridiculous, and I'd be embarrassed walking by her fundraiser booth as she asked for my help in front of everyone. For most of my life, being Filipino was more of an afterthought than the forefront of anything in my life.

Then one night, as my mom was cooking something in the kitchen, I heard her humming a tune that seemed strangely familiar.

When I asked her what it was, she said, "I used to sing it to you as a baby."

I asked her for the name of the song. The next time I saw Mara, I asked her about it.

"It's an old folk song taught to Filipino children," she said. "To teach us about how we were colonized."

The song awakened memories of the Philippines in me. I started remembering the tiny details of my childhood there.

“Can you write the lyrics for me?” I asked Mara. “In both Tagalog and English.”
When she finished and handed me the paper, I practiced singing it myself:

*Atin ku pung singsing
Metung yang timpukan
Amana ke iti
King indung ibatan.
Sangkan keng sininup
King metung a kaban
Mewala ya iti,
E ku kamalayan.*

*I once had a ring
With a beautiful gem
I inherited this
From my mother
I stored it as well as I could
In a hopebox
But it just suddenly disappeared
Without my notice.*

I think when you’ve been engrossed in a foreign country’s culture for so long, after you’ve been living the good life for years, it’s only then you realize—that’s when you start to appreciate where you’re from.

Dr. Isabelle Tremblay, Director of Donnelly Centre for Cellular and Biomolecular Research, University of Toronto:

It’s one of our oldest questions in science: nature vs. nurture. How much of a person is a result of genetic factors or environmental upbringing factors? In today’s world—at least in the Western world—we tend to have a strong sense of “self.” It’s not enough to ask *what* we are, but *who* we are. I believe the issue lies in what constitutes someone as part of a culture. Despite what most people would like to admit, how they look on the outside plays as big a role as how they sound, what they eat, and what they value on the inside.

I was there during the international summit when Elias Cruz revealed to the world that he had made the first genetically engineered human, the first designer baby. In fact, the secret was leaked a week before, and social media was already buzzing with hashtags about CRISPR

babies. Elias Cruz was already scheduled to give a presentation during the summit on his CRISPR research. But since the truth of his CRISPR son was leaked, everyone who attended the summit, the entire scientific community, along with millions of people watching the stream online, expected it to be the topic of his presentation. It was one of the most intensely anticipated and scrutinized scientific presentations in history.

The reveal was nothing short of spectacular. Elias Cruz did indeed explain what he had done, but more importantly, why. At the time, no one could really believe it. Even as he showed his slides full of experimental data, even as he talked about the exact method he used to achieve his results, there was still a looming feeling in the room that it was all some sort of grand hoax.

Before his research, germline editing in humans was still in its infancy. Although the idea of designer babies was technically possible, there were still many hurdles to overcome, many risks to mitigate, and many legal barriers to commercial implementation.

Then Elias Cruz changed everything. What most people think of as distinct genetic “traits” are more often than not affected by multiple variations of genes working together. Elias Cruz’s breakthrough was developing a method to predict with explicit certainty the outcome of these polygenic connections, so that we could edit humans with an almost God-like level of control, and they would turn out the exact way we wanted them to.

The following reaction to his presentation was immediate. A prominent researcher had snapped to his feet and shouted at everyone to arrest him, calling him an irresponsible rogue scientist, though at first no one was at his side. The summit’s schedule was changed to have Penrose-Forsyth and Liu Yanlin question him for a further forty-five minutes. A large number of attendees—politicians and scientists alike—were already calling for a global moratorium on germline editing, citing Elias Cruz’s breakthrough as “disgusting” or “inhumane” behaviour. The leading scientists of CRISPR, collectively in one room at that summit, resisted these calls, fearing it would stifle the growth of research in the field. They held a lot of influence in that regard, and they commanded the room to calm down.

After that summit, authorities were on him within 24 hours. But since then, he’s managed to evade detainment, with most people believing he is currently hiding somewhere in the Philippines, guarded by his personal armed forces.

Perhaps it would have been different if Elias Cruz had suggested his research would be used to cure diseases or help people. His true proposition—lightening skin to gain privilege—clearly echoed historical and modern systems of eugenics and racial bias, and would trigger major ethical outrage. Perhaps that is why he had calculated his retreat accordingly.

Today, there is still no global moratorium on germline editing in humans. It is a criminal offense in a large number of countries, and in others, there are strict regulatory hurdles in place, but there are still some places where bioengineers can get away with their experiments

due to poor oversight. The Philippines is purportedly one of those places, considering how easy it was for Dr. Cruz to bring his knowledge back to the country to continue his highly controversial research away from public scrutiny.

You have asked me for my personal opinion on this matter. While I do sympathize with Mara Tan, I also believe the potential benefits of CRISPR to society are unimaginable. Rejecting the technology because we are uncertain about the way it may be used would be a mistake. We've always told ourselves that it would take millions of years to evolve, but now we can do it in a single generation. Elias Cruz has killed evolution.

Aisuno Cruz, son of Elias Cruz:

[The son of Elias Cruz is a very attractive man. Some of the crew are nervous around him.]

Dad named me after a god.

He talked about it a lot. *Aisuno* was the supreme child deity of the Eskaya people, an indigenous tribe from Bohol Island in the Visayas. One of their myths tells the story of a silver bell, known as the *Lingganay nga Ugis*. This bell is said to have been stolen by a Spanish priest, but then later retrieved by Baroko, a legendary bird, who dropped it into the Kamayaan River. The bell waits for Aisuno's return.

The prophecy states that only Aisuno's arrival and retrieval of the bell will save the Eskaya people. If anyone else retrieves the bell first, it will trigger a catastrophic flood that will destroy the world. Once Aisuno retrieves the bell, the Eskaya will go through a cosmic rebirth, restore them to their true forms, and gain their "second bodies." They will be liberated from their bondage.

It's easy to see the parallels between their story and mine. That's why Dad named me Aisuno. Dad believes Filipinos have a curse and that I'm somehow going to herald a new era that will free them from it. Apparently, I will usher in a great wave of genetic editing that will wash over the entire world.

If you think about it, the story of the Philippines has always been a war against floods. The islands themselves could be thought of as the peaks of a massive underwater land only revealed by sea level, and so our land is just a mountain range covered by a flood. The islands were populated by numerous floods of peoples, from the Negritos to the Austronesians to the Spanish to the Americans. During the rainy season, flooding is common, and we are constantly fighting typhoons. The story of the Filipino people is one of survival against all odds, of clinging onto a wooden plank at sea, of hope and faith against the flood of the world.

There has always been a sentimental beauty to that. But that sentiment is fragile; ever since CRISPR was invented, we have found a gene editing tool cheaper and more precise than

anything else before, and now we are masters of our destinies. We are no longer bound by our DNA. The question of “nature vs nurture” becomes pointless, as everything is now planned, expected, chosen. My dad started a fire that humanity can’t put out.

Today, things between us are awkward. I look nothing like him, but I *want* to be his son. I’ve seen how my friends get treated by their fathers. They get taken to basketball practice, or they do *gigil*—and their fathers get an aggressive happiness from their adorableness. But my dad doesn’t treat me the same way. He treats me like an experiment, or a saviour—yet he can’t even stand looking at me. I’m an outcast in my own family. I can’t even hug Mom properly; I’m too tall. She ends up hugging my stomach.

Dad made me in his dream vision. He gave me pale skin, dirty blond hair, and striking blue eyes. He gave me immunity to genetic diseases. I can also run faster, lift more, and see farther than my friends. They stopped inviting me to basketball long ago because they think that when I play with them, it’s too unfair.

He gave me very nice traits, I’ll admit. But there was nothing wrong with him and my mom in the first place. If I were given the choice, I would give up all these traits in a heartbeat. I would rather be sick than alone.

I just want to look like them and be a part of the family. I just want to look Filipino.

Rafael Santos:

The modern world is more European than most people would like to admit. I believe we should face the facts about Western culture’s dominance around the world. The industrial revolution was started by the white man, and some of our best ideas, such as science and democracy, started in Europe. In the case of the Philippines, which was colonized by the USA, the USA is also a child of European ways of thinking, legal systems, and values. It is realistic to assume that most non-Europeans are born in this Eurocentric world and will encounter, or will be subject to, at some point in their lives, involuntary beliefs about the ways the world should work, and that since we live with this harsh reality, we must take any pragmatic solution to release us from it. CRISPR is one of those solutions that completely gets rid of the problem.

Elias Cruz has been called a white supremacist, a eugenicist, an ethnic cleanser, and many more. When I see him, I see an intellectual who only wants what’s best for his people. He also gives the choice itself, and unlike what his enemies believe, does not force his treatments on anyone. To me, he is the ultimate realist. He doesn’t solve impossible issues by pretending they are solvable; he surrenders and finds the best common ground. As a historian, I’ve studied thousands of years of history, and there have always been a few constants we have never been

able to break free from. I agree with him that it's impossible to build an ideal world without discrimination, because it would go against human nature.

Vance Tan:

Growing up, I thought America was the greatest country in the world. I remember reading about US history, and when I read about the United States' good relations with the Philippines in World War II, it made me think they were our fellow brothers. It gave me an untainted view of them, and I always thought they were saviours, especially since they liberated us from Spain. This view was reinforced by the many local war films I watched at the time, which portrayed the Americans as benevolent liberators.

It's easy to forget that they introduced themselves to us by brutal massacre.

My great-grandmother on my maternal side was a 13-year-old girl when the Philippine-American War broke out. She also witnessed the "first strike" of the *Katipuneros* from her window, the symbolic act that would launch the Philippine Revolution in 1896.

As a kid visiting her, I asked her for stories from her childhood. She described how American troops burned their houses and rice granaries, then destroyed the entire village.

Needless to say, I was stunned.

"You're confusing the Americans with the Japanese!" I interrupted. "Americans are God's great gift to the Philippines!"

My great-grandmother exploded, saying, "I am not that senile that I can't tell an American from a Japanese! It is like identifying a cow from a goat! They both did the same horrific things... just in their own language!"

She then described how some of her elder relatives took up arms during these two wars with Spain and the Americans. She was also the one who taught me that America had faked the liberation of the Philippines. As I listened to her, I was shocked to learn that after winning the war against Spain, the Americans secretly made a deal with the Spanish to buy the islands for \$20 million. As part of their agreement, they would stage a fake battle between American and Spanish troops in Manila and end with the Spanish surrendering.

"But why would they do that?" I asked her.

"Because at the time, Filipino revolutionaries were winning against the Spanish, and they were getting humiliated," she said. "They told the Americans they would rather lose to the fellow white man than to Filipinos, so they did the fake battle to make it seem like that's what happened."

When I told Mara this, she was appalled, but also thrilled to hear about a primary source for the war so close to us.

“My grandfather absolutely detested the Americans, too,” she said. “He refused to eat at any American fast-food restaurant. But my father? He was a real Amboy. He never sang the latest edition of *Lupang Hinirang*, always preferring the English version the Americans made, *Land of the Morning*. It’s unbelievable how much the public opinion of a nation can change within just one generation.”

That’s when Mara started focusing on the Philippine-American War.

GunitAI prompt #19, “Gang Rape at Engbino Household,” August 17th, 1899:

It is a hot summer evening in the street of Calle Cervantes, Manila. It is just after midnight, and American soldiers are breaking into houses of Filipino families and leaving trails of burglary and looting. They are laughing, shouting, and drunk.

Three gunshots are fired. A man, his wife, his mother, and baby are sleeping in their household when they hear them outside. Antonio Engbino jumps up and starts waking his family to take cover. Cristine Engbino quickly rushes to get the baby as Antonio helps his mother. They find cover and listen. Antonio’s mother starts praying the rosary.

They’ve never seen an *Amerikano* devil before. They know the war has been going on for a few months now, and every day, they fear they won’t live to see its end.

“Don’t go near the windows,” Antonio Engbino says to his wife.

She wraps their child in a blanket and places him under their table.

They hear laughter outside. Three American soldiers are outside their door. Their names are Privates Barringer, Sterling, and McCawley. The Engbino family hears them talk, but they cannot understand them.

“Did you see what those little creatures did to Lockhart?”

“The poor kid. They captured him and hacked him to pieces!”

“Let’s make ‘em pay.”

There is a sudden knock on the door. They are shouting more words that the Engbinos cannot understand. When no one answers the knock, they kick the door down. The three American soldiers enter the house and find the Engbino family, paralyzed in fear.

“Hey, Sterling, not bad, huh?”

“Oh, she’ll do.”

They approach Cristine Engbino. She cowers and closes her eyes. She won’t open them until hours later.

Antonio Engbino gets up, but they forbid him to do so. When he refuses to comply, they push him back and threaten him with pistols. He crouches down and covers his head with his hands, afraid they’ll shoot.

They take Cristine Engbino to the corner and throw her down. Then they rape her. They also bite her face. She is too scared to resist. Her eyes are still closed, and she knows they have guns pointing at her. After the first one finishes raping her, the other one approaches her and prevents her from getting up and rapes her.

Antonio Engbino still has his head under his hands on the ground. Getting up is risky. He can do nothing but listen to the sounds of the rape.

But at some point, the sounds become unbearable. Motivated by a furious desire to protect his wife, he screams at the top of his lungs and attacks the soldiers. He is shot twice in the chest and collapses to the ground. The gunshots wake the baby and he starts crying, but the soldiers are unfazed and continue raping his mother.

The third soldier takes her mother-in-law into the mosquito netting and rapes her.

When they finish, they open their household's trunk and throw the clothes out. They steal some and leave.

"Antonio?" says Cristine Engbino, her eyes still closed. But her husband doesn't answer her. The baby continues to wail.

Journal excerpt from Private Clifford Allred, February 6th, 1902:

I was stationed on the island of Marinduque. When you really look at it, it's not a big island. We were given orders to herd the entire population of the island into five concentration camps. We told the natives to bring whatever they could of their property, and anything they left behind would be taken by us or destroyed.

The conditions of the camp were horrendous. Everywhere smelled like rotten fish and decay. It didn't help that the other soldiers made a routine of abusing the natives and giving them fresh injuries that would get infected. We thought that some of the natives we rounded up in the camps could be secret guerrillas, so we set up boundaries and a curfew at each camp. Anyone caught outside of the boundaries after the curfew was shot on sight without hesitation. One of my officers called it a "zone of death." There were many who tried to escape, and I remember seeing their bodies spread out across the beach because nobody cared to clean them up.

The huddled population suffered and died at a rapid rate from malnutrition and disease. One in particular that troubled us a lot was dengue fever. It ravaged the natives and even some of our soldiers. Sometimes, when one of them showed strong symptoms of it, we shot them, no questions asked, then carried them out of the camp. The natives started hiding their sick family members after that, or they lied to us that they weren't sick.

After a while at the camp, I was ordered to join a platoon to set up trade blockades on other islands. We regularly burned down entire villages. In some, we were ordered to shoot as many

natives as we could; it didn't matter if they were combatants or not. If I remember correctly, I think it's because there were a few holdout rebels in the remote islands that we wanted to clear quickly, so that we could officially call the insurrection over and label all future Filipino army soldiers as "thugs." We were just given orders to kill and burn as many natives as possible we could find on those islands. The more we did, the more our generals would be pleased.

GunitAI prompt #25, "The Water Cure," September 22nd, 1902:

A high-value informant for the Filipino guerrillas has been captured by American troops. They throw him onto the muddy ground, and four soldiers pin his limbs, one for each arm and leg. A fifth soldier lifts his head slightly off the ground and uses one hand to pinch his nose.

"I've been in this nigger-fighting business for twenty-five days, and I don't want to spend one more second in this hellhole," says Colonel Leonard, the commanding officer. "The sooner we find your friends in the mountains, the sooner my boys and I can all go home. Take my advice: choose the wiser option here. You can escape what's about to happen if you just tell us where your friends are hiding. There is only one way out of this, little Indian."

The Filipino prisoner shouts back in Tagalog: *"Speaking English to me in my own country? What an asshole! Son of a whore! Eat shit!"* He spits into the face of the one pinching his nose above his head. The soldier remains stoic and wipes it off his face.

Colonel Leonard nods. "Give me the funnel."

A soldier retrieves it for him. He brings it to the prisoner's mouth. The prisoner tries to close his mouth, but his nose is pinched, and eventually, he needs to open it for air. When he does, Colonel Leonard shoves the funnel inside his mouth.

Another soldier comes forward with a gallon of water. He starts pouring it into the funnel. The prisoner is forced to swallow it, and the soldiers watch him convulse and strangle. The first gallon of water is followed by a second, then a third. By this point, the prisoner's stomach swells so large, it looks ready to burst. To his horror, they follow it with a fourth gallon, and even a fifth. He feels like he's drowning, but somehow he's still alive. Some soldiers look away, frightened of his abnormally large stomach.

In this condition, while writhing in painful agony equal to hell, speech is impossible. And so the next part is vital.

"You two." Colonel Leonard points to two soldiers. "Now."

They step forward. They kick and stomp his stomach with their boots, trying to squeeze the water out of him as fast as possible. One of them jumps on him to speed up the process. The prisoner, in unimaginable pain, expels the water from his mouth until it spills all around his clothes, coughing and spitting. Some blood comes out as well.

Colonel Leonard kneels and pulls his head up by his hair. “Ready to tell us where they are?”

The prisoner is barely conscious, but the colonel has performed this method on many natives before and knows it’s all an act. Once the water is out, the prisoner should not have a problem speaking. He gives him another minute. When the minute passes without any information, he stands up and motions for his soldiers to run the procedure again.

“We’ll be here all night if we have to,” he tells the prisoner.

They do it three more times, and then the prisoner dies.

Rafael Santos:

It is true that Americans did all those things.

The rapes, although not as systemic or widespread as the ones known in Berlin or Nanjing, definitely existed, and they spread fear among the women in Filipino communities. It is also true that entire villages were razed and entire populations were massacred for the fun of it. Most notably, we have some pictures of the piles of dead bodies of Filipino soldiers in their trenches. I’m sure Mara used them for another GunitAI prompt already. And yes, the Water Cure was particularly infamous at the time. We have multiple testimonials that say it wasn’t just used once or twice, but numerous times on many Filipinos during the war.

But let’s not forget the reason we have these accounts and stories in the first place. It might be shocking to Mara’s supporters, but many Americans at the time condemned the atrocities of the war. Most famously, the Anti-Imperialist League was created on June 15, 1898, as an effort to stop the spread of American imperialism overseas. It was specifically made to battle the annexation of the Philippines. Mark Twain was a notable vice-president of it.

My point is that many of the first-hand witness accounts we have of what actually happened during the war came from the pamphlets this organization published at the time. In a way, they were performing a role similar to Mara Tan’s. They were trying to shock the American public into standing up against the evils that went against the values of their constitution, so that they could finally end the war and the suffering of the Filipinos.

Mara Tan deliberately chose to ignore this. She also conveniently chose to ignore the atrocities Filipinos committed as well. We have accounts of Filipino soldiers cutting the ears off of American soldiers slowly so they would feel the pain, burying them alive, or burying them with their head above the ground, keeping their mouths open with a bamboo stick, and tracing lines of sugar into their mouths, letting ants do the rest. Mara Tan is just trying to make America look bad when we were just as bad, if not worse. Don’t ask me. Look at the evidence.

What other things has Mara Tan chosen to ignore in her pursuit of nationalistic Filipino propaganda? Let’s see. I’ve heard many people, including some of my academic colleagues,

argue that she conveniently left out the atrocities committed by the Japanese during World War II. Manila, in particular, suffered the most. It has been said that Manila was the most devastated city in World War II, second only to Warsaw. Some believe it was even worse than Warsaw. As a result, when I travel around the city, it amazes me to think everything I see was built after 1945. Nothing from before that survived.

Although the Japanese occupied us the shortest, they arguably committed the worst atrocities we've ever suffered in our history, even worse than anything the Spanish or Americans did. Simply calling them crimes against humanity is ludicrously insufficient. This is a common public view, and there could be many reasons for this, a likely one being the fact that the Japanese occupation is more recent and therefore the most well-documented.

Regardless, the public speculation against Mara Tan was that she wanted to create a Pan-Asian identity against colonialism, and showing the atrocities of an Asian race against another Asian race didn't help her cause. Convenient forgetfulness became her weapon. These same skeptics tried to remind her that the Chinese have also been super racist to Southeast Asians, calling them uncivilized, conquering them into their dynasties, and subjugating them and putting them under their influence. It's just the pain of recent events, of European domination, that blurred and biased her views.

The debate here is not whether the acts of the Spanish or Americans are heinous—we all know they were—but whether they were overlayed over a context of colonial supremacy, or if they were just results of just another kind of war that we've had so many of in history. Mara Tan has plenty of evidence about the things the Americans did, but she does not have any evidence to say if their actions were fuelled by the ideas of genetic superiority that she is trying to fight today. Does she actually *care* about fighting Elias Cruz and Genaris? Or is she using him as an opportunity to advance what she's always wanted to do, which is to make the whole world pity the Philippines?

Unlike her, my wife and I don't choose to pity ourselves or our country anymore. I'm here to say some good news: we've decided to move to America! In a way, I can describe the decision as one that was a long time in the making, but at the same time, it was an impromptu decision. Our children are high-school age, and they are not happy. We're doing our best to explain to them why we need to move. We are excited to have them experience a new, better life there.

Contrary to what Mara Tan probably believes, we are not going there to leave our pasts behind. I am a history teacher, and it goes against my principles. But at the same time, we *are* going for new lives. Living in the Philippines is a draining experience. Your votes don't matter, the heat is unbearable, and a natural disaster occurs every week. Saving this place is going to take much more than one person with an AI propaganda machine. That's why we're leaving.

Bahala na. It's in God's hands now. At some point, you just have to grow up and realize you can't change everything that's wrong with the world, and you just have to move on.

Aleksander Kowalczyk:

Mara has always been one of my best students. While I focused on the methodological and technical aspects of creating hyperrealistic AI videos, she was always thinking of the grand implications of their usage. Of course, I was no stranger to this topic, but we had different ways of analyzing the prospects. I always focused on the ethics, on our responsibility to protect the public from the harm that it could cause. As a professor, it was also my job. Mara was more optimistic. She believed in its power to persuade, to highlight issues with captivating empathy, and to ignite emotions to get people fully behind a cause.

To me, this only solidified the essences that separated a professor and a student. A professor does research well until they retire, but usually, after years of being hardened by it, they gain an understanding of their peak abilities and already come to terms with what they know they can do. But a student is young and naturally curious, even rebellious at times. A student is unrestrained by logic, powered by creativity, and never counts out massive but asinine ideas. A student has a naive yet optimistic view of how the world works, and they always believe, since their lives are just beginning, that they have the power—sometimes even the responsibility—to change it.

The core issue with the public is not actually the ethics of Mara's AI videos, but what in a story constitutes as *truth*. Knowing who Mara was as a person, she would never fake her sources. There is no doubt that the stories she uses are real. But depicting them properly is a whole other question. Are we really sure it was *that* horrific? Are we really sure this was the *exact* way it happened? History is not a show that must go on. For every event in history, when a certain year passes, the stage is closed by the curtains of time. Along with it goes truth, which gets lost forever.

We live on the lands of our ancestors. We walk the paths they walked, use the artifacts they used, and have rights under constitutions that they spent their lives fighting for. But when that curtain closes, our connection to their truths is gone. All we can do is find evidence at best, speculate at worst. Even recreations and adaptations in film will never capture one hundred percent of an era.

Mara must have known all this. But when Elias Cruz's CRISPR movement was taking off, she must have gambled on the public's belief in the integrity of her methods. She must have believed she had no choice. As soon as she made this decision, she ceased to be a rational computer scientist. She became a fierce activist. In that venture, I cannot support her.

Transcript from the live webcast of the First International Summit on Human Germline Editing, co-hosted by the Hong Kong Academy of Sciences (ASHK), the Royal Society (UK), the U.S. National Academy of Sciences, and the U.S. National Academy of Medicine at the University of Hong Kong, courtesy of Beijing News:

[Elias Cruz sits on a cushioned chair on the stage, with Penrose-Forsyth and Lui Yanlin sitting on both of his sides. Three small tables are in front of them with cups of water. They are all holding mics. Elias Cruz seems to hold his nervously, yet he simultaneously manages to maintain an unapologetic attitude. Penrose-Forsyth gestures to the huge audience in front of them.]

Penrose-Forsyth: As you can see, there's a big demand in this room for you to explain what you've done. I don't want you to fear what you're about to say. As long as you're in this room, no one can touch you. We've set up this session specifically so you can answer our questions.

The first question I have is about your manuscript. You said you didn't plan for the news of your son to be leaked, and that you had expected him to be revealed when *Nature* would publish your manuscript. Is that correct?

Elias Cruz: Yes. I had submitted my paper to *Nature* a few weeks earlier. I was expecting them to publish it some time before this summit, and then I would talk about it during my presentation.

Penrose-Forsyth: We have the editors of *Nature* here with us now. Do you understand that although your research is astounding, they have chosen not to publish it yet because of the significant ethical issues it imposes?

Elias Cruz: I understand.

Liu Yanlin: Your methods in your experiments are still unclear to us. There seems to be a lack of transparency in them, and many are already suggesting fraud. In your presentation, we observed that some of your data suggested your reproductive cells underwent *mosaicism*—the variation was present in some of them, but not all, implying your CRISPR edits failed to apply to all the cells of your son.

Elias Cruz: Thank you for raising these concerns. I understand that the data appears to show mosaicism in some of the reproductive cells. I want to clarify a few things that may explain what you're seeing before assuming any conclusions about my work.

First, the editing machinery may not have been delivered to *every* cell at the earliest embryonic stage. If the editing happened after the first cell division, some cells would already be unedited, which naturally yields mosaicism in the resultant reproductive cell population.

Second, there may also have been *sampling bias*. The reproductive cells we had access to for testing might not have been representative of all of them. It's possible that the subset tested

included fewer edited cells than the population average, or vice versa. Limited sample size tends to exaggerate the number of affected cells.

This procedure is only effective when the cells split and copy the DNA. Therefore, it's best on embryos, not adults. I intend this procedure to be accessible for the next generation.

My son has already been born, and my team has checked that his genetic makeup and visible appearance has matched our expectations. I can assure you that my method works.

Liu Yanlin: May I ask why you haven't shown us a picture of your son already?

Elias Cruz: Much like any father, I understand his existence is controversial, and I am doing my best to protect him from the world.

Penrose-Forsyth: For your experiments, a lot of people are also horrified that the international guidelines that many worked so hard to establish were ignored. May I ask... why did you do this? What do you intend to use designer babies for?

Elias Cruz: My native country, the Philippines, has been suffering under a colonial mentality for centuries. I wish to finally level the playing field by granting genetic privilege to the next generation of Filipino citizens, such as lighter skin.

[The whole room goes quiet. There is an unmistakable sense of tension and fear in the air.]

Penrose-Forsyth: Have you discussed this with the Filipino public?

Elias Cruz: Do we need to get everyone's permission to cure cancer? When I see a problem in my country, I do something to cure it.

Penrose-Forsyth: It is a common mistake to assume the definitive solution to problems is to use science when the current environment is often more malleable. I deeply understand your concerns, but perhaps there is another way. Are you aware that many Filipinos, and many people around the world—to put it very lightly—may be against what you are doing?

Elias Cruz: I'm willing to take the blame, and one day they'll see I'm doing them a favour.

[The room erupts.]

Aisuno Cruz:

New CRISPR Filipino babies are born with mostly Eurocentric features and grow up not knowing why they look different than their friends. There are a few exceptions who kept Filipino features, but unanimously, the male babies have more height, and the female babies have more beauty.

We've formed a community. Most of us join it when we grow up, and we all share one thing when we do: a longing to return and fit in with the rest of Filipino society.

I've made friends with some of these other CRISPR children. Many of them share stories of how sad they are that they look so different. Many even hate their parents because of it. They've started coining the term "alien" to describe what they feel like compared to unedited children.

Some of them describe how they are super uncomfortable at all the extra attention from boys and girls who think they are very attractive.

One of them told me about an experience he had in high school. When class was ending, a girl he had only chatted with twice followed him with her books in her hands. From the start, she was very flirty with him and wouldn't stop asking him questions, even when he made it clear he wanted to be left alone. One day, she unexpectedly threw herself at him and shouted that they would "have cute mixed babies and run away to America together!" This deeply disturbed him, and until today, it's his most haunting memory.

There are many different personalities in our CRISPR community, but one thing we all share in common is a belief that we are just as Filipino as unedited Filipinos. We just don't feel like it. We've started a movement to tell this to the world, but it doesn't matter. No matter what we try, people still treat us differently, and it's inescapable. I hate to say it, but it's exactly what my father predicted.

Some of us have already started applying to move to Western countries. They rationalize that they want to experience a place where they look like everybody else, a place where no one will treat them differently because of it. The irony is that discrimination is everywhere, and it is even most documented in Western countries. I know their secret, and to me, it's quite honourable. They are not trying to escape discrimination; they are trying to escape being producers of colonial mentality at home. It's tragically heroic.

I know what I'm about to say is pure speculation, but listen to me. With their good looks and intelligence, they'll probably get good partners and good jobs. They might get rich and bring some of the money back home. But to me, none of that will be a good thing. In the end, they will fulfill their parents' dreams and prove that genetic superiority exists, even if they didn't want to.

Isabelle Tremblay:

The illegal gene market boomed about 6 months after the summit at Hong Kong. I still remember when my colleague, a science ethics policymaker, contacted me and said, "Isabelle. Look up [REDACTED] on the web. You can now buy genes for your children. You can design them like a Build-A-Bear."

At the time, I didn't really understand the gravity of what was happening. Since then, the website has been taken down, but five more have taken its place. The market seemed

unstoppable, and authorities have been cracking down on it for years, but progress is slow. My only hope is that Elias Cruz kept his promise to regulate most of it so that each economic class pays what they deserve to.

[Dr. Tremblay has analyzed the market herself and has compiled a list of the globally bestselling “genes.” We have asked her to give us the list.]

Here are the Eurocentric genes that sell well:

- *Lighter skin tone:* Multiple genes affect this trait. It is caused by lower melanin production due to the variants in genes such as SLC24A5, SLC45A2, and MC1R.
- *Double eyelids:* Another polygenic trait. This trait is extremely popular, intended to mimic Western beauty standards of eye shapes. Genes such as KIF26B, CASC2, RPRIP1L, PAX1 influence it.
- *Straight or loosely wavy hair:* Genes such as EDAR and TCHH have been found to impact hair texture during development.
- *Non-black hair colours:* Variants in MC1R (especially for red hair), OCA2, and HERC2 affect the pigmentation of hair.
- *Non-brown eye colours:* Primarily controlled by OCA2 and HERC2 variants that reduce melanin in the iris.
- *Narrow nose, high cheekbones, thin lips, and a more angular jawline:* These are polygenic traits again. Genes such as PAX3 and DCHS2 influence facial structure.
- *Slim or lean body with long limbs, smooth and wrinkle-free skin, and a youthful appearance:* Genes related to collagen production and skin elasticity, such as COL1A1, play a role.

It is essential to understand that science does not care about the historic or social implications of its actualities on humans. The majority of humans, for example, have dark hair and dark eyes. It just so happened that Europeans gained traits that deviated from this standard because of the climate of the region they lived in, and then Europeans developed an imperialistic culture focused on exploiting distant lands. It is important to distinguish correlation vs. causation. These two developments do not necessarily have any association with each other.

It is also essential to understand that the majority of these purchases are from people who genuinely believe these genes will give their children better lives. Like Elias Cruz, they believe it's purely practical. When you live in a privileged state, it is impossible to judge what these people are going through. I don't support Elias Cruz, but we can never forget that these are real people he is trying to help.

That's all I have to say on this matter.

Vance Tan:

Things were getting rough for the both of us. Mara began asking for international aid. She went to Canada and requested money for some marketing campaigns, promising to use the money to highlight the best aspects of Filipino culture, so that people would focus more on the beauty of our own people rather than of foreigners.

She was a notable alumni from a Canadian university and was strongly upholding the human rights values of the UN. Naturally, the Canadian government was sympathetic to her cause. They began donating large amounts of money to Mara.

But Mara secretly used the money to bring “nicer” Filipino genes into the black market instead, aiming to combat the popularity of Elias Cruz’s European genes. She told no one, not even me. When I found out, we had a huge argument. It was the worst one we’ve ever had.

Her plan backfired completely. The international community found out, and they criticized the Canadian government for funding an illegal activity in genetic editing. The Canadian government desperately tried to emphasize that the funding was for lawful purposes only—it was Mara who broke her promises. They urged the host nation, the Philippines, to prosecute her.

We thought it was over when a Philippine government official visited our house. To our surprise, he came to inform us that there were many people on our side in the Philippine government, and that they were doing what they could to delay the prosecution. Instead, they secretly supported Mara, buying her time.

Mara was labelled an international criminal, no better than Elias Cruz.

“I don’t care,” she told me. “I just want to take him down.”

As time went by, the Filipino genes sold poorly, but the European genes sold extremely well. She worked harder than ever before on her GunitAI videos.

Even though I was always by her side, Mara believed she was on her own. That a great duty had been placed upon her, and it was her sole responsibility to silence all her doubters, to fight all her enemies at once. During the weeks after she was labelled an international criminal, the catalog of her public AI video creations contained fewer and fewer hopeful videos. Her rate of making them plummeted until eventually she stopped making them altogether, and everything that came out was negative. The negative ones also started to get worse. More disturbing.

I remember the night when we argued about the time she spent on discovering these colonial atrocities. I told her it was too much. She was only harming herself and others with everything she saw. I remember asking her, “Why are you so obsessed with them? We can take an alternate route instead. We can focus on the heart of the Filipino people. That’s what the Canadian government wanted you to do.”

Then she told me the story of William Wilberforce, the British politician who spent years of his life trying to end the Atlantic slave trade, and the tool he used that caused a public outcry in Britain, which finally generated enough empathy to abolish slavery.

“It was a cross-sectional image of a slave trade ship, with African slaves tied up and packed tightly in a cargo hold, showing how they were transported,” she told me. “They were on their backs with two feet of space above them, hundreds of them side by side on the edges of the boat, like bodies in a catacomb. They would suffer like that for weeks as they crossed the Atlantic. It was so horrific, it finally convinced the British Parliament to condemn slavery. I believe if I can just dig deeper, if only I could find the most vile, depraved acts committed by our Spanish and American colonizers—something that would *truly* be shocking—I can do something similar, and finally bring the whole world to my side.”

That night, I stayed up with her at her desk as long as I could, so that she wouldn’t think she was alone.

Narration:

One year after the gene black market opened, while she was back in Canada facing trials for the misuse of her Canadian dollar funds, nobody expected the world’s leading spokesperson against the CRISPR movement to jump off a bridge into the freezing cold waters of an Ontario river. Many of her mourners evoked eerie similarities between her death and the way Filipino Americans were thrown over Pajaro Bridge during the Watsonville Riots.

For years, her death was attributed to depression from the success of European genes over other genes on the black market. Then something truly astonishing was revealed. It was discovered that before she died, she had read a news report from a blacklisted website that had gravely disturbed her. It was a confession from an anonymous person, claiming to be a “rich, white Westerner,” who was infatuated with Filipinos, only buying the Filipino genes to groom a Filipina child from birth to fulfill his colonizer sexual fantasies. The news made headlines, shocking the entire world. It was the day after discovering this that Mara Tan left for the bridge and never returned.

Since then, investigations into the case have found no credible sources for the news report. The culprit has never been identified, likely because he doesn’t exist. No strong evidence has ever been found for the story, and it is likely the story sprouted from baseless rumours from the depths of the internet. Perhaps Mara Tan knew this. Perhaps she did not. What remains clear is that it didn’t matter if the story was true; just discovering the idea was even conceived shattered all the hope and courage she had spent years cultivating for herself and the world.

Mara Tan died because she fully believed in the worst of people. Moreover, she could not bear to live with the knowledge that she had possibly become an unknowing collaborator for

a great evil. Many Genaris supporters blamed her death on her lack of knowledge, stating that the story was questionable from the start, as kidnapping children for the despicable act was easier and cheaper. Or they blamed her lack of foresight for the consequences of her initiative in the first place.

In the week after her death, millions of people worldwide simultaneously switched their profile pictures to grayscale portraits of unedited DNA strands. Cities on five continents turned off all advertising billboards for a full day as a silent protest against the commodification of human bodies. The sales of ordinary, brown-coloured contact lenses skyrocketed, as people across the world wore them in solidarity to reject the popularity of Genaris. These viral acts of mourning and defiance briefly became the largest global protest in history.

But to Mara, it would never be enough. Seeking to give hope to Filipinos all around the world, she believed she had only succeeded in giving voices to the forces of hate, apathy, and inhumanity.

Rafael Santos:

I'm sorry to say this, but the core story Mara Tan believed in, her source of hope for her entire mission against Elias Cruz, is all one big lie.

When Ferdinand Magellan reached Cebu on April 6th, 1521, he met the local ruler, Rajah Humabon. They forged an alliance, and Humabon performed a blood ritual called a *sandugo*, while Magellan made him the first native to be baptized into the Christian faith. From there, they began a conversion and subjugation campaign to unite all the nearby barangays around the archipelago.

Most people know Lapu-Lapu and his barangay refused to submit to this new empire, but they rarely go in-depth into understanding why. Did they ever stop to wonder *why* they were the only ones who rebelled, while every other barangay in the region accepted the decree with no resistance?

The truth is that Lapu-Lapu had a personal grudge against Humabon. We have historical accounts that indicate Lapu-Lapu was willing to submit to the King of Spain, but the only thing he rejected was submitting to Humabon because of their mutual enmity. His story was one of a rivalry with another tribe, *not* a symbolic fight against colonialism. Magellan was, in essence, collateral damage.

The truth often does hurt. As a historian, I don't see events in history as black and white as the average person does. I see it as a complex, interconnected web of events and people, with shifting moralities and nuances that cloud any romantic view we have of it. Because moralities shift over time, I don't doubt it is possible that Genaris could become globally accepted.

The world is always deeper than you think. It is only people who see just the surface that get angry. This is why we need to educate them.

I have mentioned before that the modern world is Eurocentric and that we must teach people to accept this fact. But the keyword is *modern*. If you zoom out and look at the bigger picture, every day in the past was a modern day once, and it had not always been heralded by the Europeans. Empires rise and fall, certain groups conquer other groups over centuries, and that is just how history runs. No matter what culture you are a part of, there was a century when your ancestors ruled the world. Everybody takes turns.

One thing I know is that big events in history, looking in retrospect, now seem as if they could not have been stopped. This is how I feel with CRISPR. Gene editing is the future, and we must embrace it in every way possible. Some people *will* have genes more advantageous than others, and some will be left in the dust. You may not like it, I may not like it, but what can we do? Once again, I abide by my mantra: we must learn to accept the modern world.

Benedicto and Edward, Mactan Shrine rioters:

Benedicto: That's right! We tore the Magellan statue down as a tribute to the death of Mara Tan! Last night, there was a huge crowd being held back by the police standing over there, who were sent by the government. We pushed and pushed against them—*ang lakis na* crowd! When the crowd got too big, they couldn't hold us back. Some of the officers actually wanted to see it taken down too, shouting "*Tumindig para sa bayan, mga kapatid na Pilipino!*" [Translation: Stand up for the nation, Filipino brothers and sisters!]" and they joined us from the start. *Sabi pa nila*, they'd rather side with their own people.

Kase, the statue was the one thing Mara Tan hated all her life, so we thought it would serve justice to take it down. *Putang ina!*—those stupid academics doubting the Lapu-Lapu story—they're all brainwashed by European ways of studying history *pare!* What a shame!

Edward: Hoy, stop being so loud, you're scaring the reporters! *Pero* yeah, he's right. We've talked with other people here, and all of us believe that European accounts of the battle could be skewed, and they could have twisted the words of history in their favour, *na parang* Lapu-Lapu was willing to submit to Magellan, just not Humabon. We don't have any proof, and to be honest, the historians know more than we do. But just the fact that we may never know, just the fact *na* there's a small chance the opposite is true—it keeps us going. Come on, Benedicto, let's go before the police arrest us.

GunitAI prompt #XXX, “It Is Finished,” (unpublished, found in her PC’s archives):

Many people ask me what it is I truly want. I don’t know how to answer them without sounding ridiculous, so most times I put on a smile and say the things I usually do, following a script in my head about how you work and what you can do.

What I really want is for the past 500 years of the Philippines to never have happened. I want everything that I’ve read, every scene that I’ve made you depict, to disappear as if all of it never existed. Since it is something no one can give me, I don’t waste effort telling them that, or even try to make them understand. But today, you’re going to help me finally make it—or at least feel it.

I want you to generate an alternate history where the Philippines was never colonized by anyone. The people and the islands remain happy and prosperous and never experience any great suffering.

Then I want you to show me Vance’s grandparents and parents living just a bit longer. They’re with my family. We’re all together, happy.

I know you can do so much more with your creativity, if I can call it that. You’re free to do anything to make these things happen in your video, as long as you get it done. Thanks for helping me all these years. I’m going to bring the final result with me and watch it as I leave. I’ll hold onto it so no one else ever sees it. This will be your final task.

Gia Reyes:

I was never into history. As a kid, I couldn’t care less about it.

Every treaty was meaningless, every battle just another date to memorize for the test. I was more interested in the future, in the research into technologies such as AI and quantum computers. For the first time ever, CRISPR, a technology meant to carry us into the future, forced us to look at our past.

Every beginning demands the ruin of what came before.

Because my family moved here when I was very young, I don’t have many memories of the Philippines. All I have are the firsthand accounts from my mom and dad. But even though I had my parents, I was conditioned to think that reminiscing was useless. Just like many other Filipino immigrant families, my siblings and I were taught there was a reason we moved here—and that we would never move back.

Growing up, whenever a discussion about the Philippines came up, all my parents talked about was the poverty, the corruption, and the constant floods. I remember the routine they would always follow whenever one of them would accidentally describe the Philippines as “home”:

Mom: "I miss those Purefoods hot dogs back home. *Sayang*, you can't get them here."

Dad: "*This* is home now."

Mom: *Curses*. "You're right."

Eventually, the few core memories I did have in my early childhood in the Philippines seemed to align with their tainted view of it. I remembered the crowded, chaotic streets of Manila, with dirty slums, jungles of power cables, and men on scooters smoking cigarettes or drinking cola in plastic bags with straws. I remember throwing tantrums at my poor *yayas* because they didn't make my fried chicken the way I liked it. I remember one particular memory where it was typhoon season, and our driver brought us home into the flooded parking lot of our apartment, and my mom had to carry me above water that went up to her knees to get to the stairs. My memories, aligned with what my parents said, all combined to embed into my mind one thing; this was a *bad* place to be, and it was a privilege to get out.

Because of that, I was always fearful to tell people I was Filipino, because maybe they would think I was from a "poor" nation. This fear eventually turned into a quiet resentment of my past. Looking back, I don't think it was extreme; I didn't hate myself like other Filipinos did. I accepted I was Filipino and answered that I was if people asked what my background was. But that was about the extent of my tolerance. For the most part, whether deliberately or not, I disassociated with being Filipino. It wasn't that I had an agenda to assimilate into Canadian culture as fluidly as possible, but more that I was indifferent to my heritage and only cared about the future, not the past.

But everything changed when I met Mara.

"Oh, *kabayan!*" she said.

I was stunned. "Excuse me?"

"Fellow countryman," she explained. "*Kamusta!* You're Filipino, right?"

"Uh, yes."

"Do you know how to speak Tagalog?"

I felt embarrassed. "No, sorry."

She giggled. "It's okay, it doesn't matter to me. I don't see many Filipinos here. Seeing another Filipino abroad is like spotting a heartbeat from home."

Today, I now understand the heartbeat of what Mara wanted to preserve. If I did not look Filipino, she would not have recognized me in that moment. Who knows what would have happened then? Perhaps we would never have met. This is what Elias Cruz wants to take away for Filipinos around the world.

Mara liked to talk about the history of the Philippines.

"*Mano po*, where you place your forehead to an elder's hand when greeting them, is sometimes said to be of Spanish colonial influence, but the gesture is actually common in Malaysia and Indonesia, called *salim*," she said.

The way she saw it, the Philippines had a heritage more in common with Southeast Asia, China, and the island of Taiwan. We could adopt Western values and lighten our skin all we want, but at our core, we shared a history with Asia.

Mara taught me that multiple prehistoric peoples from Asia settled the archipelago thousands of years ago, but they either diminished, mixed with the local population, or went extinct. Most modern Filipinos today trace their ancestry to a group that migrated from the island of Taiwan over 4000 years ago. They were called the Austronesians. They were an expert seafaring race that had every member of each boat contribute to survival, forging our spirit of familial cooperation till this day. Once they discovered the Philippine archipelago, they became masters of land as well, cultivating beautiful rice terraces that followed the contours of mountains and engineering complex irrigation systems to water them while other civilizations were still figuring out the wheel. It's hard to imagine how much impact their culture had on the linguistic and cultural foundations of the Filipino people today.

Centuries ago, we were once a very proud and powerful people. We were the richest region in Southeast Asia, with powerful kingdoms fuelled by maritime trading with the Chinese, Indians, and Arabs. Merchants got rich, and the islands were prosperous and peaceful. Now, we have lost all that. Mara told me she personally believed the current poverty and corruption of the Philippines are due to the long-lasting effects of colonialism. That the foreign powers that subjugated us are to blame, and while modern Filipinos are also to blame for the state of our country, in the end, their colonial mentality is still a result of foreigners.

At the time, these were all things I didn't know—things purposely kept from the historical revisionism of our colonists. I started doing some research of my own, and suddenly my personal beliefs about history started to transform.

When I was in school, I used to see treaties and acts of the past as irrelevant pieces of information or interesting facts that would only serve you well in trivia. I didn't know *anything* about them, and would mostly just memorize their names for tests. I would see something, such as the "Tydings–McDuffie Act of 1934," and think, "Okay, two white guy names, an act, and it happened in 1934." As a kid, I wasn't able to grasp the cultural, geopolitical, and socioeconomic significance of the people who *died* that led to this act. I didn't understand the years of abuse, torture, and deaths of the Filipinos who fought hard for the very moment this act was signed, or all the wars, councils, and factors that came into this act's existence. To me, it was just a piece of paper. To others, it was their *life's work*. That is what the curtain of time does to history, and it is why we are responsible for ensuring it never closes.

It's why I'm trying to do something about it now. Inspired by Mara, I took a Filipino history class, which covered precolonial, colonial, and postcolonial history. I felt an immense amount of pride when I learned the story of Lapu-Lapu and our first-ever victory against the Spanish. I asked my dad about my grandparents' lives as chicken farmers, then their parents' lives as

rice farmers, then their parents' lives as merchants. I learned more about the revolts against the Spanish and the brutal war with the Americans. I cried at the atrocities committed by the Japanese on the people of Manila during World War II.

For the first time ever, I understood where I came from, the things my people went through, and the shared suffering and courage that united us. Suddenly, more things in my life made sense. I was able to see the inner forces that weaved Filipino communities together, the origins of our values and core beliefs about the world, how we respected ancestors, and why.

Although I haven't seen her in a long time, this was the final gift Mara gave to me. She made me proud to be Filipino, and for that, I'll always thank her.

Vance Tan:

When I was young, my *a-má* and *kong-kong* took me on a trip to Baguio. I was so young at the time, I don't remember much about it. Today, I took a bus here, just to feel what I once did in my childhood.

Baguio is known as the "summer capital" of the Philippines. Many people go here during the summer because of its higher elevation and cooler weather, both high-profile politicians and regular citizens alike. On the day our family was supposed to go, I got terrible diarrhea, and my mother thought we should cancel. But my *a-má* insisted that we still go, and so we went. When I hear that story today, it makes me laugh. My *a-má* was such a stubborn lady.

I remember doing boat rides on Burnham Lake, a man-made pond in the middle of the city. We also ate at the famous Chinese Filipino restaurant, Luisa's Café, on Session Road. My father said it was "a place everyone who visits Baguio should eat at." Some of my older cousins went ghost-hunting in the ruins of the Hyatt Terraces Hotel. It was destroyed by an earthquake in the 90's, killing dozens of people inside, and was never rebuilt. Now tourists visit the ruins to see if they can get a glimpse of the paranormal.

When I got older, I learned a bit more about the history of Baguio. The Americans had established a military camp here in the early 1900s for rest and recreation. They established several facilities in and around Baguio, and these had strong effects on the city's growth, infrastructure, identity, and function. When they built a road to connect it to Manila, it became more accessible to the public. My *kong-kong* said, "The Americans build a base there, and suddenly all the Filipinos flock to it. It feels like many things we consider great about our country were built by foreigners. It's pathetic."

It was much different back then. My father said that ever since SM built a supermall in Baguio in 2003, the city has become heavily commercialized. He believes it has lost all of its old magic. Like me, this was his feeling of "home," and now he has lost it.

[The crew makes way as Vance pushes a wheelchair with his father, against the backdrop of the beautiful Baguio City.]

Mara used to help me take care of my father. These days, it's just us two.

Starting a few years ago, my father got a neurological condition where he can't move his muscles as easily. Despite his condition, he requested that he be brought here today. He's as stubborn as his mother, my *a-má*. Returning here was always his goal before he died.

My father was a renowned scholar before his condition began. Once it started, he deteriorated rapidly. Right now, his thoughts are too slow to perceive anything. He insisted on writing one last letter to me before he knew he would reach this state.

I know you reporters are looking for personal stories for your film, but that you'd never ask anyone to share something unless they offered. It's OK, you can read this letter.

[Vance Tan becomes emotional and requests that we do not film him any longer. He gives the letter, translated into English as best he could, and gives us special permission to add it here.]

Son,

I am sorry that Mara, your wife and my beloved daughter-in-law, has gone on to Heaven.

She was truly a special person, and her creation was just as special. I have no doubt that because of her, even after she is gone, the world will refuse to forget the memories of our people.

No one thinks this, but GunitAI is the same as us. It has its own genetics, which is the neural network that Mara designed for it. But what it creates completely depends on what it is trained on. It is like raising a child. Give it bad things, it will spout bad things. Give it good things, it will spout good. GunitAI's creations are a result of how the model was designed, but also a culmination of everything it has seen. We have a duty to raise our kids the right way, so that in the future, they will believe the right things.

That being said, I hope I raised you well.

Son, I have to tell you something.

I was born with a rare, life-altering genetic disease. The doctors told me it is caused by a mutation in a single gene, and its effects only show later in life. When they show, you lose control of movement in your muscles, your limbs become stiff, and you get problems with memory.

What is different for everyone is when the disease appears. Luckily, it only appeared when I got old. One day, I spent ten minutes trying to find a word to describe something, and it was then that I realized that my thinking was slowing down.

Your mother and I kept this a secret from you because we didn't want you to fear your future the way I did.

Before that moment, I lived my entire life in fear. Every day, I was afraid I would wake up to find that I had lost control of my muscles, and my life would be over. This fear made me live every day as if it were my last, and I ended up taking risks I wouldn't have otherwise. One of them was choosing to court your mother on the first day I met her. Two years later, you were born.

In the end, I overcame my genetic disease and the fear associated with it, and everything turned out for the better. There is always the possibility of passing the disease onto children, but even if God decided that the disease would appear in my twenties, I would fight tooth and nail against anyone who decided I shouldn't give birth to you. Look at how our family turned out in the end. I'm so proud of you.

Son, I think the world changing is inevitable. There is nothing we can do about it. One day, we celebrate the past, the next we invent tools that erase it. But I want you not to give up hope. As humans, we have the ability to make rules and regulations and work through things together. Making decisions purely based on protecting the dignity of people is one of the greatest things about us.

We are terrified of the idea of 'genetically superior' people. But the truth is, we already select babies for medical conditions. Mutations are only abnormal in a statistical sense, for every species will have them. But if there is even a slight suspicion of a genetic disease, of which are also mutations, we abort them without hesitation.

Being alive is truly a gift, no matter the condition of the body you experience it in. Sometimes when I am sad, I wonder, if CRISPR was a thing when I was born, would I have been allowed to exist?

Narration:

We are at the Mara Rosario Tan memorial monument in Manila. Two years ago, this shrine was requested by the Filipino people and erected by the Philippine government, and now it embodies hope as a worldwide symbol. On the front plaque, in both Tagalog and English, it mentions that the monument is not just to honour Mara Tan, but also stands as a defiant response against the horrific ideologies that plague our past and our species as a whole. Every day, the monument stands as a shining beacon against foreign subjugation, notions of genetic superiority, and the evil theories that go against all of humanity.

“Feel the love of a million ancestors,” the plaque reads.

Ever since CRISPR technology was invented, humanity has picked a road that it can’t go back. Almost unanimously, the world has agreed to enforce a suspension on gene editing in humans. We are still not sure what to do with a world where parents can now choose the skin colour, personality, and intelligence of their children. The human genome, once a silent poem, can now be edited with clarity and purpose. Now comes the era where we separate flesh and spirit, where we erase physical scars etched deep in ancestral and cultural memory. The possibilities are endless and terrifying. For now, we wait until we learn what they truly are.

Mara Tan was a lover of her own family, culture, and history. To her, she cherished the idea that she came from a long line of people who had endured centuries of atrocities and cultural erasure to ultimately come out on top with their own identity. To live in a world where none of that mattered, where all of that could be taken away from you in the blink of an eye, was an unbearable nightmare for her. She was fully connected to her past, but when CRISPR came out, she became haunted by the idea of losing it. She died believing Elias Cruz had won, that the thousands of years of human history had all been sacrificed for an ideology that failed to celebrate the beauty of human life.

But Mara Tan forgot that the story of humanity has never been about our genes. No matter what we look like, we are shaped by forces beyond genetics, and we adapt to the people and environment around us. This is what makes us truly who we are. Every day, we are surrounded by invisible ties to our homelands, creating our belief systems and core outlooks on life. Each word we speak is molded by generations of ancestors who spoke it until it reached us in its final form today. Each idea we hold descends from the ancestors who built the community that allowed us to think the idea in the first place. Just by being born, your name gets uttered, your mass affects the rotation of the Earth, and the lives of the people who meet you are changed forever. None of these is determined by genes.

Elias Cruz was half right. When you look at a person, you are seeing the final result of millions of generations mixing and shuffling genes. But at the same time, you are witnessing the final product of thousands of years of culture, language, food, thoughts, and love.

These things will never go away, unless we choose to let them. Now is the time to advocate for restrictions on changing what it means to be human, to defend the preservation of culture, and to remember all those who suffered before us that gave us our lives today. We cannot hide their memories behind the curtain of time. We must learn all that we can about where we come from; it's the only way to understand who we are.

Author's notes

This story is dedicated to Filipinos around the world and everyone who fought hard for the Filipino identity.

The documentary format of this story was inspired by Ken Liu's "The Man Who Ended History: A Documentary."

The following sources were consulted for the research of this story. I am deeply grateful to all those who studied the topics of this story, especially those who investigated Filipino history and psychology for the Filipino people. Any errors or misinterpretations of their works are entirely my own.

Main sources:

Francia, Luis H. 2010. *A History of the Philippines: From Indios Bravos to Filipinos*. New York: The Overlook Press.

Davies, Kevin. 2020. *Editing Humanity: The CRISPR Revolution and the New Era of Genome Editing*. New York: Pegasus Books.

All other people who directly or indirectly aided in the creation of this work:

Xiao Chua (contemporary Filipino historian), Tony Olaes (founder of The Filipino Story organization), Renato Constatino (Filipino historian), Virgilio G. Enriquez ("Father" of Filipino psychology), Red Chua (Filipino American YouTuber), Stuart Creighton Miller (American historian and educator of U.S. imperialism and Asian American history), Henry T. Greely (American lawyer specializing in biomedical technology ethics), Siddhartha Mukherjee (Indian American physician, biologist, and author), Melanie Mitchell (American computer scientist), and various other Filipino article writers, national heroes, activists, university students, and internet commenters who shared their personal stories.

For the phrase "the love of a million ancestors":

Heins, Camelia. 2022. "The Effects of Colonial Mentality Are Long-Lasting on Filipino Youths." *New University*. January 13, 2022. <https://newuniversity.org/2022/01/13/the-effects-of-colonial-mentality-are-long-lasting-on-filipino-youths/>.

To learn more about the Watsonville Riots:

Showalter, Michael P. 1989. "The Watsonville Anti-Filipino Riot of 1930: A Reconsideration of Fermín Tobera's Murder." *Southern California Quarterly* 71 (4): 341–48. <https://doi.org/10.2307/41171455>

Public Affairs. 2021. "Preserving the Legacy of Watsonville's First Filipino Immigrants." *UCSC News*, May 4, 2021. <https://news.ucsc.edu/2021/05/manong-generation-oral-history/>

For the descriptions of rape on Filipinas during the Philippine-American War:

Tera Kimberly Maxwell, *Imperial Remains: Memories of the United States' Occupation of the Philippines* (PhD diss., University of Texas at Austin, 2011)

Final notes:

The GunitAI prompts were based on real stories from the Philippine-American War, from letters published by the Anti-Imperialist League.

Parts of the GunitAI videos of Mary Winter's great-grandfather were written by an actual AI, to simulate their "creativity."

The title of Mara's last GunitAI prompt, "It Is Finished," was inspired by José Rizal's last words before his execution under the Spanish.

The summit where Elias Cruz presented his research is modelled after the Second International Summit on Human Genome Editing, which took place from November 27-29, 2018, at the University of Hong Kong, co-hosted by the same organizations that were mentioned.

The character Elias Cruz was inspired by He Jiankui, the first scientist to attempt to edit the human genome of two fetuses, twins Lulu and Nana. At the time of this writing, they are still alive somewhere today.